MATH 423: Lecture 3

Risky Assets

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Probability: Basic definitions

- Let $(\Omega, \mathbb{P})$ be a discrete probability space. A random variable (RV) X is a mapping $X : \Omega \to \mathbb{R}$. E.g., $\Omega = \{\omega_1, \omega_2, \omega_3\}$ with probabilities

$$\mathbb{P}(\omega_1) = \frac{1}{2}, \quad \mathbb{P}(\omega_2) = \frac{1}{3}, \quad \mathbb{P}(\omega_3) = \frac{1}{6}.$$

  Set the RV X by

$$X(\omega_1) = 0, \quad X(\omega_2) = 0, \quad X(\omega_3) = 6.$$

- The expectation of X is defined as

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega)\mathbb{P}(\omega) = \sum_{k \in X(\Omega)} k\mathbb{P}(X = k).$$

  In the last example

$$\mathbb{E}[X] = 0 \cdot \frac{1}{2} + 0 \cdot \frac{1}{3} + 6 \cdot \frac{1}{6} = 0 \cdot \left(\frac{1}{2} + \frac{1}{3}\right) + 6 \cdot \frac{1}{6} = 1.$$

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Probability: Basic definitions

- The variance of an RV X is given by

$$\text{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \geq 0,$$

where the last inequality holds if and only if X is a constant, in which case it equals its expectation. In the previous example

$$\text{Var}(X) = (0-1)^2 \cdot \frac{1}{2} + (0-1)^2 \cdot \frac{1}{3} + (6-1)^2 \cdot \frac{1}{6} = 5.$$

The standard deviation (SD) of X is $\sigma_X = \sqrt{\text{Var}(X)}$.

- The covariance of RVs X and Y is given by

$$\begin{aligned}\sigma_{XY} = \text{cov}(X,Y) &= \mathbb{E}\left[(X - \mathbb{E}[X])\,(Y - \mathbb{E}[Y])\right] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].\end{aligned}$$

Obviously, $\text{cov}(X,X) = \text{Var}(X)$. Also, if X and Y are independent then $\text{cov}(X,Y) = 0$ (which we refer as uncorrelated). The other direction is typically wrong. However, in the normal (Gaussian) case it is true.

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Probability: Basic definitions

- The correlation coefficient of RVs X and Y is given by

$$\rho_{XY} = \rho(X, Y) = \frac{\sigma_{XY}}{\sigma_X \sigma_Y} \in [-1, 1],$$

where

$$\begin{aligned} \rho_{XY} &= +1 \qquad \text{iff} \qquad Y = aX + b, \quad a > 0; \\ \rho_{XY} &= -1 \qquad \text{iff} \qquad Y = aX + b, \quad a < 0. \end{aligned}$$

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Probability: Properties

- (Linearity of Expectation) Let X and Y be RVs, and $a, b \in \mathbb{R}$. Then,

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

- (Bilinearity of Covariance) Let $\{X_i\}_{i=1}^n$ and $\{Y_j\}_{j=1}^m$ be RVs, and $\{a_i\}_{i=1}^n$ and $\{b_j\}_{j=1}^m$ be scalars. Then,

$$\text{cov}\Big(\sum_{i=1}^n a_i X_i, \sum_{j=1}^m b_j Y_j\Big) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j \text{cov}(X_i, Y_j).$$

- (Variance of linear combination) Let X and Y be RVs, and $a, b \in \mathbb{R}$. Then,

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + 2ab\text{cov}(X, Y) + b^2\text{Var}(Y).$$

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Risky assets: Dynamics of stock prices

We denote the stock price at time t by $S(t)$ for any $t \geq 0$.

In particular, $S(0)$ stands for today's stock price, which is known to everyone in the market. The future stock price $S(t)$ is random.

Mathematically, $S(t)$ is defined as a positive r.v. on a probability space Ω consisting of all feasible price movement "scenarios" $\omega \in \Omega$.

When we want to be more precise, we shall write $S(t, \omega)$ to denote the stock price at time t if the market follows the scenario $\omega \in \Omega$.

Now, suppose that time runs in a discrete manner: $t = n\tau$, where $n = 0, 1, 2, \ldots$ and τ is a fixed time step. The step τ can be any time unit, a year, a day, a minute and etc.

For simplicity: we write $S(n)$ instead of $S(n\tau)$ for $n = 0, 1, 2, \ldots$

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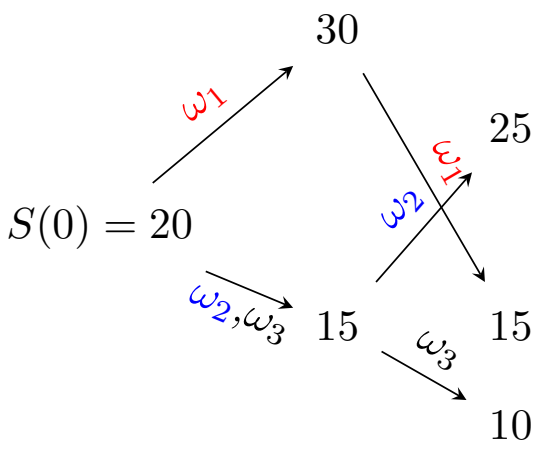
Eg 3.1

Suppose there exist three possible market scenarios, $\Omega = \{\omega_1, \omega_2, \omega_3\}$ s.t.

Scenario	$S(0)$	$S(1)$	$S(2)$
ω_1	20	30	15
ω_2	20	15	25
ω_3	20	15	10

We can use a tree structure to represent price movements:



As we note, today's stock price $S(0)$ is the same for all scenarios (why?).

Thus there is only one node on the leftmost; it is called the root (of the tree).

We can identify each scenario with a path of the tree from the root through to one of the leaves on the rightmost.

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Return

For two integers $n < m$, we substitute $K(n, m)$ for $K(n\tau, m\tau; \omega)$ as the return over the time interval $[n\tau, m\tau]$ when the scenario ω is realized:

$$K(n, m) := K(n\tau, m\tau) = \frac{S(m\tau, \omega) - S(n\tau, \omega)}{S(n\tau, \omega)} = \frac{S(m) - S(n)}{S(n)}.$$

The return is a random variable as well.

We can rearrange this equality so that

$$S(m) = S(n)(1 + K(n, m)).$$

Here we see that growth factor $= 1+$ return.

Analogously, we can simply write $k(n\tau, m\tau; \omega)$ by $k(n, m)$ for the log return over time interval $[n\tau, m\tau]$:

$$k(n, m) := \ln \frac{S(m)}{S(n)}.$$

We can also rearrange this equality to get

$$S(m) = S(n)e^{k(n,m)}.$$

Here growth factor $= \exp$(log return).

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Return

Question: What happens when the stock pays a dividend of div(n) at time $n\tau$?

Like coupon being paid in a coupon bond, when a dividend is paid, the stock price drops by the same amount.

Thus $S(n) + \text{div}(n)$ reflects the stock price at time $n\tau$ if the dividend were not paid. So the one-step return and log return are

$$K(n) := K(n-1, n) = \frac{S(n) + \text{div}(n) - S(n-1)}{S(n-1)},$$

and

$$k(n) := k(n-1, n) = \ln \frac{S(n) + \text{div}(n)}{S(n-1)}.$$

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Prop 3.1

If no dividends are paid, then

$$1 + K(n, m) = (1 + K(n+1))(1 + K(n+2)) \cdots (1 + K(m)).$$

Moreover, if one-step returns $K(n+1), \ldots, K(m)$ are independent, then

$$1 + \mathbb{E}[K(n, m)] = (1 + \mathbb{E}[K(n+1)])(1 + \mathbb{E}[K(n+2)]) \cdots (1 + \mathbb{E}[K(m)]).$$

1st statement:

$$LHS = \frac{S(m)}{S(n)},$$

$$RHS = \frac{S(n+1)}{S(n)} \frac{S(n+2)}{S(n+1)} \cdots \frac{S(m)}{S(m-1)}.$$

2nd statement: use the independence property: when X, Y are independent, we have

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

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Prop 3.2

If no dividends are paid, then

$$k(n,m) = k(n+1) + k(n+2) + \cdots + k(m).$$

Moreover,

$$\mathbb{E}[k(n,m)] = \mathbb{E}[k(n+1)] + \mathbb{E}[k(n+2)] + \cdots + \mathbb{E}[k(m)].$$

Note: no assumption on independence is needed.

1st statement:

$$LHS = \ln \frac{S(m)}{S(n)},$$

$$RHS = \ln \frac{S(n+1)}{S(n)} + \ln \frac{S(n+2)}{S(n+1)} + \cdots + \ln \frac{S(m)}{S(m-1)}.$$

2nd statement: for two random variables X, Y,

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

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Eg 3.2

Suppose that $K(1), K(2), K(3)$ are i.i.d. (independent and identically distributed). If the expected return $\mathbb{E}[K(0,2)] = 10\%$, find $\mathbb{E}[K(1)]$ and $\mathbb{E}[K(0,3)]$?

By Prop 3.1,

$$1 + \mathbb{E}[K(0,2)] \overset{\text{indep.}}{=} \big(1 + \mathbb{E}[K(1)]\big) \cdot \big(1 + \mathbb{E}[K(2)]\big) \overset{\text{ident.}}{=} \big(1 + \mathbb{E}[K(1)]\big)^2$$

$$\Rightarrow \quad \mathbb{E}[K(1)] = \sqrt{1 + 10\%} - 1 \approx 0.0488.$$

Therefore,

$$\begin{aligned} 1 + \mathbb{E}[K(0,3)] &\overset{\text{indep.}}{=} \big(1 + \mathbb{E}[K(1)]\big) \cdot \big(1 + \mathbb{E}[K(2)]\big) \cdot \big(1 + \mathbb{E}[K(3)]\big) \\ &\overset{\text{ident.}}{=} \big(1 + \mathbb{E}[K(1)]\big)^3 \approx 1.15369 \end{aligned}$$

$$\Rightarrow \quad \mathbb{E}[K(0,3)] = 0.15369.$$

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Binomial tree model

For stock prices, we extend One-Step Binomial Model to multiple steps so as to obtain a binomial tree structure for stock prices.

Assumption 1 (A1)

One-step returns $K(n)$, $n = 1, 2, \ldots$ are i.i.d. such that for each n,

$$K(n) = \begin{cases} u, & \text{with prob } p, \\ d, & \text{with prob } 1-p, \end{cases}$$

where $-1 < d < u$ (why?) and $p \in (0, 1)$.

☼ $K(n) = \frac{S(n)-S(n-1)}{S(n-1)} > \frac{-S(n-1)}{S(n-1)} = -1.$

Assumption 2 (A2)

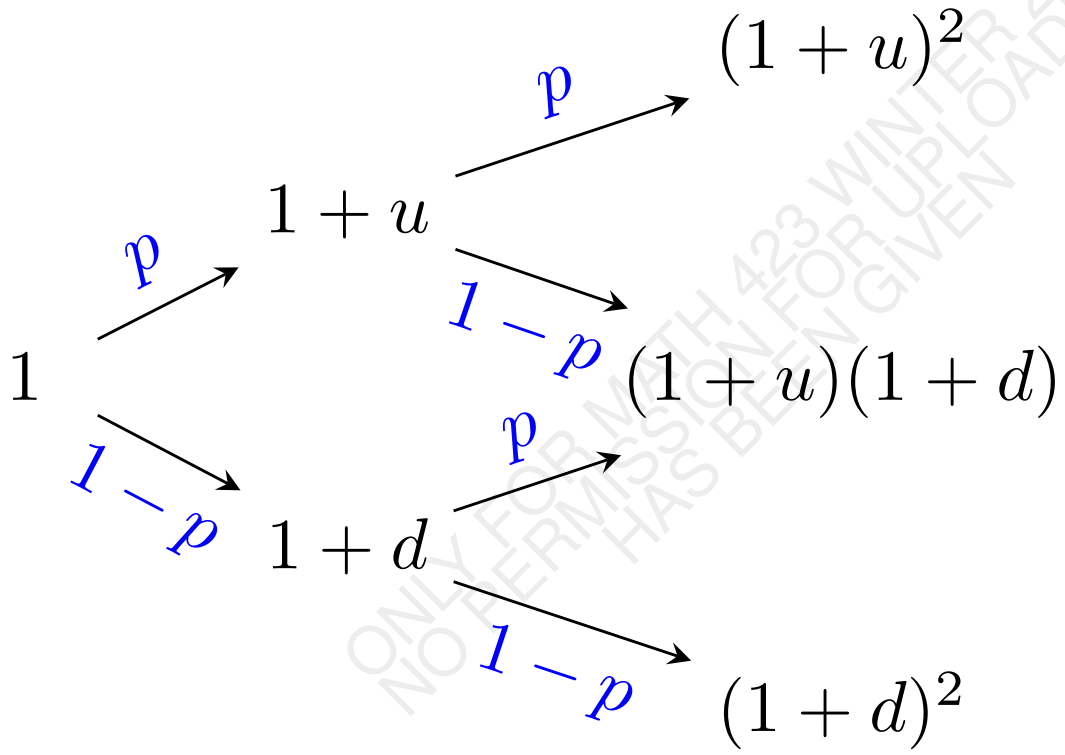
The one step return r on a risk-free investment (bonds or cash held in a bank account) is the same at each time step and $d < r < u$ (why?).

☼ By no arbitrage principle (Ex 1.1),

$$\frac{S^d}{S(0)} < \frac{A(1)}{A(0)} < \frac{S^u}{S(0)} \quad \Rightarrow \quad 1 + d < 1 + r < 1 + u \quad \Rightarrow \quad d < r < u.$$

Binomial tree model

Assume for simplicity that $S(0) = 1$. The process for stock prices looks a tree: each node except the leaves has two branches. Starting from the root, each time going up (or down), the stock price obtains a factor $1 + u$ (or $1 + d$).



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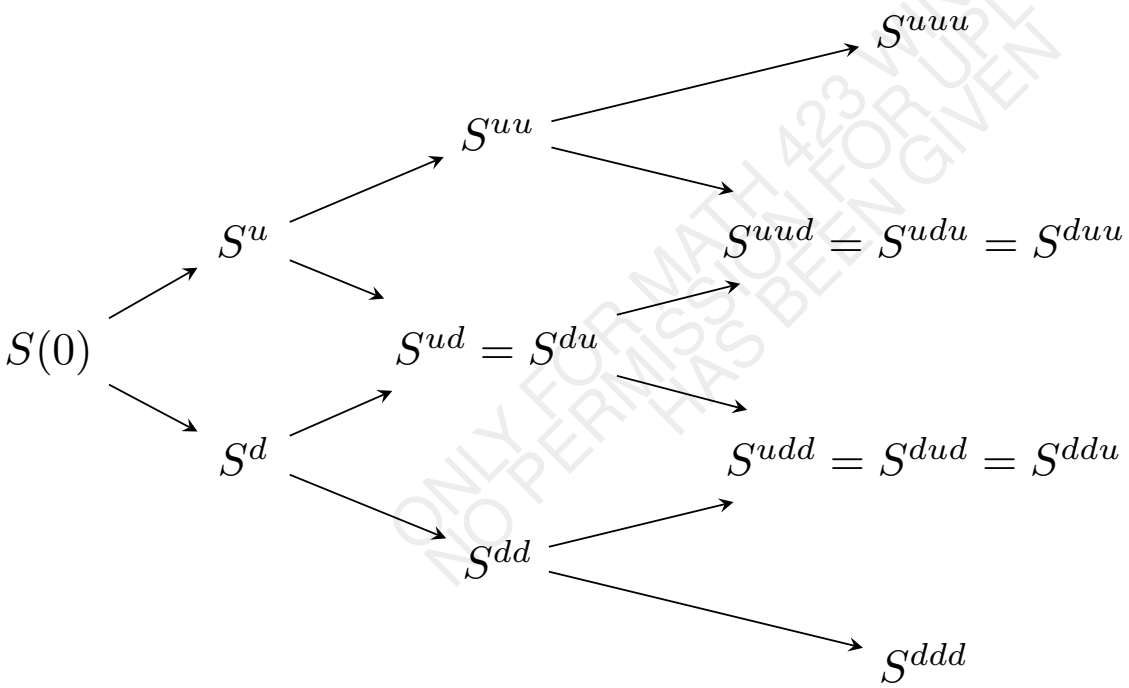
Binomial tree model

In general,

$$S(n) = S(0) * (1+u)^i * (1+d)^{n-i}, \text{ for } i = 0, 1, \ldots, n,$$

i.e. $S(n)$ can take $n+1$ different values. The obtained tree is also called the recombining tree.



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Binomial tree model

Question: Given $i \in \{0, 1, \ldots, n\}$, what is the probability that

$$S(n) = S(0) * (1+u)^i * (1+d)^{n-i}.$$

$$\begin{aligned}
&\mathbb{P}(S(n) = S(0) * (1+u)^i * (1+d)^{n-i}) \\
&= \mathbb{P}(\text{the movement has } i \text{ ups and } n-i \text{ downs}) \\
&= \binom{n}{i} p^i (1-p)^{n-i}.
\end{aligned}$$

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